

ITA1412 -ETHICAL HACKING

Experiment no:2

Experiment Name: Host Discovery

Procedure:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering- >select Nmap)

Step 2: Perform different types of scans

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

To perform host discovery

-Pn only port scan nmap -Pn192.168.1.1

-sn only host discover nmap -sn192.168.1.1

-PR arp discovery on a local network nmap -PR192.168.1.1

-n disable DNS resolution nmap -n 192.168.1.1

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~ x kali@kali: ~ x  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcr2-10.46 libpcap-1.10.6 nmap-libdnet-  
1.18.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 84878sec preferred_lft 84878sec  
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
        valid_lft 86291sec preferred_lft 14291sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$ nmap -sn 10.0.2.15/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:10 -0400  
Nmap scan report for 10.0.2.2  
Host is up (0.00038s latency).  
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)  
Nmap scan report for 10.0.2.3  
Host is up (0.0012s latency).  
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)  
Nmap scan report for 10.0.2.15  
Host is up.  
Nmap done: 256 IP addresses (3 hosts up) scanned in 7.91 seconds  
(kali@kali)-[~]  
$ ping 10.0.2.2  
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.  
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=1.32 ms  
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=1.28 ms  
64 bytes from 10.0.2.2: icmp_seq=3 ttl=64 time=0.777 ms  
64 bytes from 10.0.2.2: icmp_seq=4 ttl=64 time=2.34 ms  
64 bytes from 10.0.2.2: icmp_seq=5 ttl=64 time=1.19 ms
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(kali㉿kali)-[~]  
$ ping 10.0.2.2  
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.  
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=1.32 ms  
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=1.28 ms  
64 bytes from 10.0.2.2: icmp_seq=3 ttl=64 time=0.777 ms  
64 bytes from 10.0.2.2: icmp_seq=4 ttl=64 time=2.34 ms  
64 bytes from 10.0.2.2: icmp_seq=5 ttl=64 time=1.19 ms  
^C  
— 10.0.2.2 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4026ms  
rtt min/avg/max/mdev = 0.777/1.379/2.337/0.516 ms
```

```
(kali㉿kali)-[~]  
$ hostname  
kali
```

```
(kali㉿kali)-[~]  
$ netstat -rn  
Kernel IP routing table  
Destination      Gateway         Genmask         Flags         MSS Window  irtt Iface  
0.0.0.0          10.0.2.2       0.0.0.0         UG            0 0        0 eth0  
10.0.2.0         0.0.0.0        255.255.255.0   U            0 0        0 eth0
```